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### Cart-based availability determination for an online concierge system

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#### Abstract

An online concierge system may use images received from shopping carts within retailers to determine the availability of items within those retailers. A shopping cart includes externally-facing cameras that automatically capture images of the area around the shopping cart as the shopping cart travels through a retailer. The online concierge system receives these images, which depict displays within the retailers from which a picker or a retailer patron can collect items. The online concierge system determines which items should be depicted in the images and which items are actually depicted in the images. The online concierge system identifies which items should be depicted, but are not depicted, and determines that these items are unavailable (e.g., out of stock) at that retailer. The online concierge system updates an availability database to indicate that these items are unavailable and may notify the retailer that the item is unavailable.

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**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of co-pending Patent Cooperation Treaty (PCT) International Application No. PCT/CN2022/111574 filed on Aug. 10, 2022, with the China National Intellectual Property Office as the Receiving Office, which is herein incorporated by reference.

## BACKGROUND

(1) An online concierge system is an online system that allows a user to order items from a retailer for delivery to the user. An online concierge system may be operated by a third party that coordinates pickers to collect items from retailer locations to service user orders, but the online concierge system may not be operated by the retailers themselves. As such, the online concierge system may lack access to inventory and availability data for items offered for sale at different retailer locations. The online concierge system can request this data from retailers, but the data may not be formatted such that the online concierge system can easily use it. Furthermore, different retailers may record different information when tracking their inventory. For example, one retailer may record inventory on a location-by-location basis, whereas another retailer may record their inventory based on regions within which their locations are situated. Thus, even if retailers were willing to provide their inventory data to the online concierge system, the online concierge system may require complex systems to ensure that the data from each retailer is consistent and can be used. Thus, online concierge systems may require significant computing resources to generate availability data for different retailers and different locations.

## SUMMARY

(2) In accordance with one or more aspects of the disclosure, an online concierge system may use images received from shopping carts within retailer locations to determine the availability of items within those retailer locations. A shopping cart includes externally-facing cameras that automatically capture images of the area around the shopping cart as the shopping cart travels through a retailer location. The online concierge system receives these images from these shopping carts. These images depict displays within the retailer locations from which a picker or a retailer patron can collect items. The online concierge system determines which items should be depicted in the images and which items are actually depicted in the images. The online concierge system identifies which items should be depicted, but are not depicted, and determines that these items are unavailable (e.g., out of stock) at that retailer location. The online concierge system updates an availability database to indicate that these items are unavailable and may notify the retailer that the item is unavailable. The online concierge system also may have a picker verify that the items are actually unavailable within the retailer location.

(3) The online concierge system uses the availability database to keep track of which items are available and adjusts the presentation of items to users accordingly. For example, a customer may transmit a search query to the online concierge system requesting that the online concierge system provide items that relate to the search query. The online concierge system may generate a set of candidate items, but may check the availability database to ensure that these items are available. If an item is not available, the online concierge system may filter that item out from the set of items sent to the user's client device for presentation to the user. The online concierge system also may still transmit the unavailable item to the user, but with instructions that the unavailable item be presented with an indication that the item is unavailable.

(4) By determining the availability of items based on images captured by a shopping cart, the online concierge system does not need to reformat or interpret data from independent systems, thereby reducing the computational load on the online concierge system. Additionally, by reducing or eliminating the requirement for human interaction to determine the availability of items, the online concierge system reduces the operational costs of creating an up-to-date availability database for items offered by a retailer.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 illustrates the environment of an online concierge system, in accordance with some embodiments.
- (2) FIG. 2 is a block diagram of an online concierge system, in accordance with some embodiments.
- (3) FIG. 3A is a block diagram of the customer mobile application (CMA), according to some embodiments.
- (4) FIG. 3B is a block diagram of the picker mobile application (PMA), according to some embodiments.
- (5) FIG. 3C is a block diagram of the runner mobile application (RMA), according to some embodiments.
- (6) FIG. 4 is a flowchart for an example method for determining the availability of an item in a retailer based on an image captured from a shopping cart, in accordance with some embodiments.
- (7) FIG. 5 is an example image captured by a shopping cart, in accordance with some embodiments.

### DETAILED DESCRIPTION

#### (8) Environment of an Online Concierge System

- (9) FIG. 1 illustrates an example system environment **100** of an online concierge system **102**, according to some embodiments. The system environment **100** illustrated in FIG. 1 includes an online concierge system **102**, a customer **104**, a customer mobile application **106**, a picker **108**, one or more retailers **110**, a picker mobile application **112**, a runner **114** and a runner mobile application **116**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 1, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.
- (10) Additionally, the figures use like reference numerals to identify like elements. A letter after a reference numeral, such as “**110a**,” indicates that the text refers specifically to the element having that particular reference numeral. A reference numeral in the text without a following letter, such as “**110**,” refers to any or all of the elements in the figures bearing that reference numeral. For example, “**110**” in the text refers to reference numerals “**110a**” and/or “**110b**” in the figures.
- (11) The environment **100** includes an online concierge system **102**. The online concierge system **102** is configured to receive orders from one or more customers **104** (only one is shown for the sake of simplicity). An order specifies a list of goods or items to be delivered to the customer **104**. Goods, items, and products may be used synonymously herein to mean any item that a customer **104** can purchase via the online concierge system. The order also specifies the location to which the goods are to be delivered, and a time window during which the goods should be delivered. In some embodiments, the order specifies one or more retailers from which the selected items should be purchased. The customer may use a customer mobile application (CMA) **106** to place the order; the CMA **106** is configured to communicate with the online concierge system **102**.
- (12) The online concierge system **102** is configured to transmit orders received from customers **104** to one or more pickers **108**. A picker **108** may be a contractor, employee, or other person (or entity) who is enabled to fulfill orders received by the online concierge system **102**. The environment **100** also includes three retailers **110a**, **110b**, and **110c** (only three are shown for the sake of simplicity; the environment could include any number of retailers). The retailers **110** may be physical retailers, such as grocery stores, discount stores, department stores, etc., or non-public warehouses storing items that can be collected and delivered to customers. As used herein, a “retailer” is an entity that

operates “retailer locations,” which are physical buildings or stores in which the retailer operates. Each picker **108** fulfills an order received from the online concierge system **102** at one or more retailers **110**, delivers the order to the customer **104**, or performs both fulfillment and delivery. In some embodiments, pickers **108** make use of a picker mobile application **112** which is configured to interact with the online concierge system **102**.

(13) The shopping cart **120** is a vessel that a user can use to hold items as the user travels through a retailer location. The shopping cart **120** includes one or more cameras **130** that capture image data of the shopping cart's storage area and a user interface **140** that the user can use to interact with the shopping cart **120**. The shopping cart **120** may include additional components not pictured in FIG. **1**, such as processors, computer-readable media, power sources (e.g., batteries), network adapters, or sensors (e.g., load sensors, thermometers, proximity sensors).

(14) The cameras **130** capture image data of the shopping cart's storage area. The cameras **130** may include internally-facing cameras **130a** that capture two-dimensional or three-dimensional images of the shopping cart's contents. These cameras **130s** are coupled to the shopping cart **100** such that the cameras **130a** capture image data of the storage area from different perspectives. Thus, items in the shopping cart **120** are less likely to be overlapping in all camera perspectives. In some embodiments, the cameras **130** include embedded processing capabilities to process image data captured by the cameras **130**. For example, the cameras **130** may be MIPI cameras.

(15) The shopping cart **100** may include one or more sensors that capture measurements describing the shopping cart **100**, items in the shopping cart's storage area, or the area around the shopping cart **100**. For example, the shopping cart **100** may include load sensors that measure the weight of items placed in the shopping cart's storage area. Similarly, the shopping cart **100** may include proximity sensors that capture measurements for detecting when an item is being added to the shopping cart **100**. The shopping cart **100** may transmit data from the one or more sensors to the online concierge system **102**.

(16) In some embodiments, the shopping cart **120** captures a set of image data in response to detecting that an item is being added to the storage area. The shopping cart **120** may detect that an item is being added based on sensor data from sensors on the shopping cart **120**. For example, the shopping cart **120** may detect that a new item has been added when the shopping cart **120** detects a change in the overall weight of the contents of the storage area based on load data from load sensors. Similarly, the shopping cart **120** may detect that a new item is being added based on proximity data from proximity sensors indicating that something is approaching the storage area of the shopping cart **120**. The shopping cart **120** captures image data within a timeframe near when the shopping cart **120** detects a new item. For example, the shopping cart **120** may activate the cameras **130** and store image data in response to detecting that an item is being added to the shopping cart **120** and for some period of time after that detection.

(17) The cameras **130** may include cameras **130b** that are facing externally to the shopping cart **120**, meaning that the cameras **130b** capture images of regions around the shopping cart **120**. Each externally-facing camera **130b** has a different field of view of a region around the shopping cart **120**. The externally-facing cameras **130b** may be oriented such that the fields of externally-facing cameras cover most or all of the area around the shopping cart **12**.

(18) In some embodiments, the externally-facing cameras **130b** are oriented such that the externally-facing cameras **130b** capture images of displays within a retailer location as the shopping cart **120** travels through the retailer location. Displays are structures within the retailer location for presenting items to patrons or pickers within the retailer location. For example, displays include shelves, tables, platforms, cabinets, cases, clothes racks, or boxes.

(19) The shopping cart **120** automatically captures images using the externally-facing cameras **130b** as a picker or a patron uses the shopping cart **120** to collect items in the retailer location. The shopping cart **120** may capture images using the externally-facing cameras **130b** at regular intervals or in response to a trigger. For example, the externally-facing cameras **130b** may capture images

when the shopping cart **120** detects that the shopping cart **120** is moving or when the shopping cart **120** reaches certain locations in the retailer location. The shopping cart **120** transmits images captured by the externally-facing cameras **130b** to the online concierge system **102**.

(20) The shopping cart **120** may include an indoor location system that determines the location of the shopping cart **120** within the retailer location. The indoor location system uses sensor data from sensors on the shopping cart **120** or within the retailer location to determine the location of the shopping cart. For example, the indoor location system may use GPS data from a GPS tracker coupled to the shopping cart, or may use RFID tags positioned within the retailer location to determine the location of the shopping cart **120**. In some embodiments, the indoor location system uses wheel rotation data to determine the location of the shopping cart **120** within the retailer location. An example indoor location system for a shopping cart **120** is described in PCT Application No. PCT/CN2022/102796, entitled “Shopping Cart Self-Tracking in an Indoor Environment” and filed Jun. 30, 2022, which is incorporated by reference. The indoor location system regularly determines the location of the shopping cart **120**, and transmits the shopping cart's location to the online concierge system **102**.

(21) The shopping cart **120** uses the indoor location system to determine the location of the shopping cart **120** at the time when the externally-facing cameras **130b** capture images of the area around the shopping cart **120**. For example, the shopping cart **120** may trigger the externally-facing cameras **130b** to capture images of the area around the shopping cart **120** whenever the indoor location system determines the location of the shopping cart **120** (or vice versa).

(22) The shopping cart **120** includes a user interface **140** through which the user can interact with the online concierge system **130b**. The user interface **140** may include a display, a speaker, a microphone, a keypad, or a payment system (e.g., a credit card reader). The user interface **140** may allow the user to adjust the items in their shopping list or to provide payment information for a checkout process. Additionally, the user interface **140** may display a map of the retailer location indicating where items are located within the retailer location. In some embodiments, a user may interact with the user interface **140** to search for items within the retailer location, and the user interface **140** may provide a real-time navigation interface for the user to travel from their current location to an item within the retailer location. The user interface **140** also may display additional content to a user, such as suggested recipes or items for purchase.

(23) Online Concierge System

(24) FIG. 2 is a block diagram of an online concierge system **102**, according to some embodiments. The online concierge system illustrated in FIG. 2 includes an inventory management engine **202**, an inventory database **204**, an order fulfillment engine **206**, a transaction records database **208**, a picker management engine **210**, a picker database **212**, a customer database **214**, a communications engine **216**, and an availability determination module **218**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 2, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

(25) The online concierge system **102** includes an inventory management engine **202**, which interacts with inventory systems associated with each retailer **110**. In some embodiments, the inventory management engine **202** requests and receives inventory information maintained by the retailer **110**. The inventory of each retailer **110** is unique and may change over time. The inventory management engine **202** monitors changes in inventory for each participating retailer **110**. The inventory management engine **202** is also configured to store inventory records in an inventory database **204**. The inventory database **204** may store information in separate records—one for each participating retailer **110**—or may consolidate or combine inventory information into a unified record. Inventory information includes both qualitative and quantitative information about items, including size, color, weight, SKU, serial number, and so on. In some embodiments, the inventory

database **204** also stores purchasing rules associated with each item, if they exist. For example, age-restricted items such as alcohol and tobacco are flagged accordingly in the inventory database **204**. (26) In some embodiments, the inventory database **204** stores item embeddings for items offered for sale on the online concierge system **102**. An item embedding is an embedding that describes an item. The item embeddings may be associated with specific items stored by the inventory database **204**. For example, each brand of an item may have an individual item embedding, or items may have different item embeddings for each retailer that sells the item. Alternatively, each item embedding may be associated with a generic item, and each generic item is associated with specific items that are similar or substitutes of each other. For example, the inventory database **102** may store an item embedding for the generic item “milk”, and the specific items of “Moo Moo 2% Milk” and “Greener Pastures Organic Whole Milk” may both be associated with the item embedding for “milk.”

(27) The inventory database **204** includes an availability database that stores data describing the availability of items. An item is available if the item can be collected by a picker at a retailer location, and an item is unavailable if the item cannot be collected by a picker. For example, an item may be unavailable if the item is out-of-stock at a retailer location, meaning the retailer location does not have the item available for purchase to the public. In some embodiments, an item is unavailable if the item is not present on displays within the retailer location on which the item should be displayed for collection by customers or pickers. The availability database may store the availability data for an item on a per-location basis. For example, the availability database may indicate that an item is available at one retailer location and unavailable at another. In some embodiments, the availability database stores an availability score indicating the likelihood that an item is available at a retailer location for each item-location pairing.

(28) The online concierge system **102** uses the availability database to determine which items to present to a customer. For example, if a customer searches for an item, the online concierge system **102** checks the availability database to determine whether the item is available. If the item is available, the online concierge system **102** presents the item to the customer. If the item is unavailable, the online concierge system **102** may not present the item to the user in search results or may indicate to the user that the item is unavailable.

(29) The availability database is generated and updated by the availability determination module **218**. The availability determination module **218** receives images from shopping carts **120** and determines whether items are unavailable based on whether items that are expected to be depicted in the image data are actually depicted. The availability determination module **218** determines a set of items that the availability determination module **218** expects to detect in an image from the shopping cart **120**. The availability determination module **218** determines which items should be detected in the image based on the location of the shopping cart **120** in the retailer location. The availability determination module **218** compares the expected set of items to a set of items that the availability determination module **218** actually identifies in the image, and determines which of the expected set of items were not identified in the image. Those items are determined to be unavailable at the retailer location within which the shopping cart **120** is located and the availability determination module **218** updates the availability database to indicate that those items are unavailable. The availability determination module **218** and the availability database, in accordance with some embodiments, are described in further detail below.

(30) The online concierge system **102** also includes an order fulfillment engine **206** which is configured to synthesize and display an ordering interface to each customer **104** (for example, via the customer mobile application **106**). The order fulfillment engine **206** is also configured to access the inventory database **204** in order to determine which items are available at which retailers **110**. The order fulfillment engine **206** determines a sale price for each item ordered by a customer **104**. Prices set by the order fulfillment engine **206** may or may not be identical to in-store prices determined by retailers (which is the price that customers **104** and pickers **108** would pay at

retailers). The order fulfillment engine **206** also facilitates transactions associated with each order. In some embodiments, the order fulfillment engine **206** charges a payment instrument associated with a customer **104** when he/she places an order. The order fulfillment engine **206** may transmit payment information to an external payment gateway or payment processor. The order fulfillment engine **206** stores payment and transactional information associated with each order in a transaction records database **208**.

(31) In some embodiments, the order fulfillment engine **206** also shares order details with retailer **110**. For example, after successful fulfillment of an order, the order fulfillment engine **206** may transmit a summary of the order to the appropriate retailer. The summary may indicate the items purchased, the total value of the items, and in some cases, an identity of the picker **108** and customer **104** associated with the transaction. In some embodiments, the order fulfillment engine **206** pushes transaction and/or order details asynchronously to retailer systems. This may be accomplished via use of webhooks, which enable programmatic or system-driven transmission of information between web applications. In another embodiment, retailer systems may be configured to periodically poll the order fulfillment engine **206**, which provides detail of all orders which have been processed since the last request.

(32) The order fulfillment engine **206** may interact with a picker management engine **210**, which manages communication with and utilization of pickers **108**. In some embodiments, the picker management engine **210** receives a new order from the order fulfillment engine **206**. The picker management engine **210** identifies the appropriate retailer **110** to fulfill the order based on one or more parameters, such as the contents of the order, the inventory of the retailers, and the proximity to the delivery location. The picker management engine **210** then identifies one or more appropriate pickers **108** to fulfill the order based on one or more parameters, such as the pickers' proximity to the appropriate retailer **110** (and/or to the customer **104**), his/her familiarity level with that particular retailer **110**, and so on. Additionally, the picker management engine **210** accesses a picker database **212** which stores information describing each picker **108**, such as his/her name, rating, previous shopping history, and so on. The picker management engine **210** transmits the list of items in the order to the picker **108** via the picker mobile application **112**. The picker database **212** may also store data describing the sequence in which the pickers' picked the items in their assigned orders.

(33) As part of fulfilling an order, the order fulfillment engine **206** and/or picker management engine **210** may access a customer database **214** which stores information describing each customer. This information could include each customer's name, address, gender, shopping preferences, favorite items, stored payment instruments, and so on.

(34) The online concierge system **102** may use a communication engine **216** that transmits information between the customer mobile application **106**, the picker mobile application **112**, and the runner mobile application **116**. The information may be sent in the form of messages, such as texts or emails, or notifications via application, among other forms of communication. The communication engine **216** may receive information from each application about the status of an order, the location of a customer in transit, issues with items in an order, and the like. The communication engine **216** determines a message or notification to send to a customer **104**, picker **108**, or runner **114** based on this information and transmits the notifications to the appropriate application. For example, the online concierge system **102** may receive information from the customer mobile application **106** indicating that a customer **104** is traveling to the pickup location to retrieve an order. Based on this information, the communication engine **216** sends a notification to the runner mobile application **116** indicating that the customer **104** associated with a specific order is in transit, which may incite the runner **114** to retrieve the order for pick up. In another example, the online concierge system **102** may receive a message from the picker mobile application **112** that an item of an order is not available. The communication engine **216** may transmit the message to the customer mobile application **106** associated with the order.



(35) FIG. 3A is a block diagram of the customer mobile application (CMA) **106**, according to some embodiments. The CMA **106** illustrated in FIG. 3A includes an ordering interface engine **302**, a system communication interface **304**, a preferences management interface **306**, and a location data module **308**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 1, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

(36) The customer **104** accesses the CMA **106** via a client device, such as a mobile phone, tablet, laptop, or desktop computer. The CMA **106** may be accessed through an app running on the client device or through a website accessed in a browser. The CMA **106** includes an ordering interface **302**, which provides an interactive interface with which the customer **104** can browse through and select items and place an order. The ordering interface **302** also may include a selected items list that specifies the amounts and prices of items that the customer **104** has selected to order. The customer **104** may review the selected items list and place an order based on the selected items list. Furthermore, the ordering interface **302** may present recipes to the customer **104** that the online concierge system **102** predicts the customer **104** is attempting to complete, and may provide an option to the customer **104** to add additional items needed to complete a recipe to the customer's selected items list.

(37) Customers **104** may also use the ordering interface **302** to message with pickers **108** and runners **114** and receive notifications regarding the status of their orders. The CMA **106** also includes a system communication interface **304** which, among other functions, receives inventory information from the online concierge system **102** and transmits order and location information to the online concierge system **102**. The CMA **106** also includes a preferences management interface **306** which allows the customer **104** to manage basic information associated with his/her account, such as his/her home address and payment instruments. The preferences management interface **306** may also allow the customer to manage other details such as his/her favorite or preferred retailers **110**, preferred handoff times, special instructions for handoff, and so on.

(38) The CMA **106** also includes a location data module **308**. The location data module **308** may access and store location data related to a client device associated with a customer **104** via the customer mobile application **106**. Location data may include the geographic location of the client device associated with the customer mobile application **106**, how fast the client device is traveling, the average speed of the client device when in transit, the direction of travel of the client device, the route the customer **104** is taking to a pickup location, current traffic data near the pickup location, and the like. For simplicity, the location of a customer client device or client device may be referred to as the location of the customer throughout this description. The customer **104** may specify whether or not to share this location data with the customer mobile application **106** via the preferences management interface **306**. If a customer **104** does not allow the customer mobile application **106** to access their location data, the location data module **308** may not access any location data for the customer **104**. In some embodiments, the customer **104** may specify certain scenarios when the location data module **308** may receive location data, such as when the customer **104** is using the customer mobile application **106**, any time, or when the customer **104** turns on location tracking in the customer mobile application **106** via an icon. The customer **104** may also specify which location data the location data module **308** may retrieve, and which location data is off-limits. In some embodiments, the location data module **308** may be tracking the customer's **104** location as a background process while the CMA **106** is in use. In other embodiments, the CMA **106** may use real-time location data from the location data module **308** to display a map to the customer **104** indicating their current location and the route to a pickup location for their order.

(39) FIG. 3B is a block diagram of the picker mobile application (PMA) **112**, according to some embodiments. The PMA **112** illustrated in FIG. 3B includes a barcode scanning module **320**, a

basket manager **322**, a system communication interface **324**, and an image encoder **326**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 1, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

(40) The picker **108** accesses the PMA **112** via a mobile client device, such as a mobile phone or tablet. The PMA **112** may be accessed through an app running on the mobile client device or through a website accessed in a browser. The PMA **112** includes a barcode scanning module **320** which allows a picker **108** to scan an item at a retailer location **110** (such as a can of soup on the shelf at a grocery store). The barcode scanning module **320** may also include an interface which allows the picker **108** to manually enter information describing an item (such as its serial number, SKU, quantity and/or weight) if a barcode is not available to be scanned. The PMA **112** also includes a basket manager **322** which maintains a running record of items collected by the picker **108** for purchase at a retailer **110**. This running record of items is commonly known as a “basket”. In some embodiments, the barcode scanning module **320** transmits information describing each item (such as its cost, quantity, weight, etc.) to the basket manager **322**, which updates its basket accordingly. The PMA **112** also includes an image encoder **326** which encodes the contents of a basket into an image. For example, the image encoder **326** may encode a basket of goods (with an identification of each item) into a quick-response (QR) code which can then be scanned by an employee of the retailer **110** at check-out.

(41) The PMA **112** also includes a system communication interface **324**, which interacts with the online concierge system **102**. For example, the system communication interface **324** receives information from the online concierge system **102** about the items of an order, such as when a customer updates an order to include more or less items. The system communication interface may receive notifications and messages from the online concierge system **102** indicating information about an order. The system communication interface transmits notifications and messages to be displayed via a user interface of the mobile device associated with the PMA **112**.

(42) FIG. 3C is a block diagram of the runner mobile application (RMA) **116**, according to some embodiments. The RMA **116** illustrated in FIG. 3C includes a bag interface engine **328**, a bag manager **330**, and a system communication interface **332**. Alternative embodiments may include more, fewer, or different components from those illustrated in FIG. 1, and the functionality of each component may be divided between the components differently from the description below. Additionally, each component may perform their respective functionalities in response to a request from a human, or automatically without human intervention.

(43) The runner **114** accesses the RMA **116** via a client device, such as a mobile phone, tablet, laptop, or desktop computer. The RMA **116** may be accessed through an app running on the client device or through a website accessed in a browser. The RMA **116** includes bag interface engine **328**, which provides an interactive interface with which the runner **114** can view orders they need to deliver and the locations of the bags for each order, such as on a particular shelf or in a refrigerator of a pickup location. The runner **114** may receive notifications through the bag interface engine **328** about new orders, the location of a customer **104** who is in transit to a pickup location, and new orders to deliver. The runner **114** may also receive communications via the bag interface engine **328** with customers regarding order handoff and pickup confirmation and may interact with the interface generated by the bag interface engine **328** to send communications to customers and the online concierge system **102** regarding order status. For example, a runner **114** may send the customer a pickup spot at the pickup location to meet for order handoff and indicate that an order has been delivered to a customer via the interface, which ends the wait time calculation by the location data module **308** associated with the customer.

(44) The RMA **116** includes a bag manager **330** that manages the assignment of orders to runners **114** and the locations of bags for each order. The RMA **116** also includes a system communication

interface **332** which, among other functions, receives inventory information from the online concierge system **102** and transmits order and bag information to the online concierge system **102**. The system communication interface may also receive notifications and messages from the online concierge system **102** indicating information about an order. The system communication interface transmits notifications and messages to be displayed via a user interface of the mobile device associated with the RMA **116**.

#### Example Availability Detection Module

(45) FIG. **4** is a flowchart of an example method for determining the availability of an item in a retailer location based on an image captured from a shopping cart, in accordance with some embodiments. Alternative embodiments may include more, fewer, or different steps from those illustrated in FIG. **4** and the steps may be performed in a different order from that illustrated in FIG. **4**. Additionally, this method may be performed automatically by an online concierge system or may be performed in response to human instructions. In some embodiments, the method is performed by an availability determination module (e.g., availability determination module **218**) of an online concierge system (e.g., online concierge system **102**).

(46) The online concierge system receives **400** an image from a shopping cart (e.g., shopping cart **120**) in a retailer location. The image is captured by a camera coupled to the shopping cart, and the image depicts items on a display within the retailer location.

(47) The online concierge system determines **410** a location of the shopping cart within the retailer location at the time that the image was taken. The online concierge system may request the location of the shopping cart from an indoor location system operating on the shopping cart. Alternatively, the online concierge system may determine the location of the shopping cart based on the received image. For example, the online concierge system may identify a location marker within the image that indicates a location of the shopping cart within the retailer location. The location marker is a real-world object that is detectable by the online concierge system. For example, a QR code or a barcode may be used as a location marker. In some embodiments, the online concierge system uses barcodes for price tags for items as location markers within the retailer location. The online concierge system stores a mapping of location markers to locations within the retailer location. In some embodiments, the online concierge system identifies multiple location markers within the received image and uses the locations corresponding to each of the location markers to determine the location of the shopping cart when the image was captured.

(48) In some embodiments, the shopping cart's location includes the shopping cart's orientation within the retailer location. The online concierge system may receive the shopping cart's orientation from the shopping cart or may determine the shopping cart's orientation directly based on sensor data received from the shopping cart (e.g., based on compass data, gyroscopic data, or accelerometer data).

(49) The online concierge system identifies **420** an expected set of items based on the shopping cart's location. The expected set of items is a set of items that the online concierge system expects to be depicted by the image if all of the items were available in the retailer location. In other words, if the image depicts a display in the retailer location where an item should be displayed, then the online concierge system would identify that item as being in the expected set of items. The online concierge system may identify the expected set of items using an item map of the retailer location. The item map identifies the locations within the retailer location where items are displayed for collection by pickers or by patrons at the retailer location. The item map stores item locations in a two-dimensional plane representing the floor of the retailer location. The item map can also store item locations in a third dimension. For example, the item map may indicate on which level of a set of shelves an item is displayed. In some embodiments, the item map is a planogram specifying where items should be displayed in the retailer location. The online concierge system uses the shopping cart's location and the item map to determine which items are located within the field of view corresponding to the image. In some embodiments, the online concierge system also uses the

shopping cart's orientation to identify the expected set of items.

(50) In some embodiments, the online concierge system identifies the expected set of items based on barcodes that the online concierge system identifies in the image. For example, the online concierge system may store a machine-learning model that has been trained to identify barcodes that are coupled to displays for items. The online concierge system may use this machine-learning model to identify barcodes that correspond to items that should be displayed in the image to identify the expected set of items.

(51) The online concierge system identifies **430** an actual set of items depicted in the image. The actual set of items are the items that are present on the displays depicted in the image, meaning the items are available for collection by picker or by patrons at the retailer location. The online concierge system identifies the actual set of items by applying computer-vision techniques to the image. For example, the online concierge system may apply a machine-learning model (e.g., a neural network) to the image. This machine-learning model may be trained to identify items in an image. In some embodiments, this machine-learning model is trained to identify items by identifying barcodes depicted in the image.

(52) The online concierge system identifies **440** a difference set of items based on the expected set of items and the actual set of items. The difference set of items is the set of items that are in the expected set of items but not in the actual set of items. Thus, the difference set of items represents the items that the online concierge system expects to be depicted in the image, but that are not depicted.

(53) The online concierge system determines that the items in the difference set of items are unavailable at the retailer location and updates **450** an availability database to indicate that the items in the difference set of items are unavailable. The online concierge system may update the availability database such that the items are designated as unavailable at the retailer location within which the shopping cart is located.

(54) In addition to updating the availability database, the online concierge system may notify the retailer that the item is unavailable. For example, the online concierge system may transmit a notification to a computing system operated by the retailer that notifies the retailer that the item is out of stock. Additionally, the online concierge system may instruct a picker to verify that an item in the difference set of items is actually unavailable. For example, if the shopping cart is in use by a picker, the online concierge system may transmit a notification to the shopping cart to present to the picker that requests that the picker verify that an item is unavailable. Alternatively, the online concierge system may ask another picker that is in the retailer location to verify that the item is unavailable while that picker collects items for an order.

(55) The online concierge system uses the availability database when presenting items to a customer for the customer's selection for ordering. For example, the online concierge system may receive a search query from a customer for a set of items that are available for ordering from a retailer. The online concierge system may generate a set of candidate items that satisfy the customer's search query (e.g., using a machine-learning model) and uses the availability database to determine whether those candidate items are available. If the availability database indicates that a candidate item is unavailable, the online concierge system may filter out that candidate item from the set of candidate items, and may present only the filtered set of candidate items to the customer. Alternatively, the online concierge system presents the full set of candidate items to the customer, but indicates to the customer which of the candidate items, if any, were determined to be unavailable.

(56) In some embodiments, the online concierge system receives an order from a customer for delivery to the customer at a delivery location during a later time frame. Between the time when the customer placed the order and the later time frame, the online concierge system may determine that an item in the order is unavailable. The online concierge system notifies the customer that an item in their order is no longer available and may request that the customer select alternatives to the

unavailable item.

(57) In some embodiments, the online concierge system uses images received from the shopping cart to automatically generate labeled training data for training an item availability model. An item availability model is a machine-learning model that is trained to predict the availability of an item based on order data describing orders placed by customers, inventory data describing the inventory of items at retailers, or other suitable data that describes factors that may impact the availability of an item. The online concierge system may use the item availability model to predict whether an item is available at a retailer location, and may compare the prediction from the item availability model to an entry in the availability database that was updated based on images received from a shopping cart. In some embodiments, the online concierge system generates training examples to which the item availability model can be applied, and which include a label generated based on images from the shopping cart.

(58) While FIG. 4 is described in the context of receiving a single image from a shopping cart, alternative embodiments may use multiple images captured by externally-facing cameras on the shopping cart to determine the availability of items at the retailer location. For example, the online concierge system may identify the expected set of items based on the shopping cart's location, and may identify the actual set of items based on items that the online concierge system detects in each of a set of images received from the shopping cart.

(59) FIG. 5 illustrates an example image **500** captured by a shopping cart, in accordance with some embodiments. The image **500** depicts a display **510** containing items **520** that are available for collection by pickers or patrons of the retailer. In FIG. 5, an item **530** is out of stock on the display **510**. The online concierge system identifies an expected set of items that includes the in-stock items **520** and the out-of-stock item **530**, and identifies the actual set of items that includes the in-stock items **520** but not the out-of-stock item **530**. The online concierge system thereby identifies a difference set of items that includes the out-of-stock item **530** and updates an availability database to indicate that the item **530** is unavailable.

(60) Additionally, the image **500** depicts potential location markers that the online concierge system may use to determine the location of the shopping cart when the image **500** was captured. For example, the online concierge system may identify barcodes **540** for the items and compare the barcodes to an item map to determine the location of the shopping cart. The online concierge system also may identify a QR code **550** that indicates the location of the display **510** in the retailer, and the online concierge system may use the location of the display **510** to determine the location of the shopping cart.

#### OTHER CONSIDERATIONS

(61) The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the patent rights to the precise pages disclosed. Many modifications and variations are possible in light of the above disclosure.

(62) Some portions of this description describe the embodiments in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

(63) Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In some embodiments, a software module is implemented with a computer program product comprising one or more computer-readable media containing computer program code or

instructions, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described. In some embodiments, a computer-readable medium comprises one or more computer-readable media that, individually or together, comprise instructions that, when executed by one or more processors, cause the one or more processors to perform, individually or together, the steps of the instructions stored on the one or more computer-readable media. Similarly, a processor comprises one or more processors or processing units that, individually or together, perform the steps of instructions stored on a computer-readable medium.

(64) Embodiments may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise a computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

(65) Embodiments may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

(66) The description herein may describe processes and systems that use machine learning models in the performance of their described functionalities. A “machine learning model,” as used herein, comprises one or more machine learning models that perform the described functionality. Machine learning models may be stored on one or more computer-readable media with a set of weights. These weights are parameters used by the machine learning model to transform input data received by the model into output data. The weights may be generated through a training process, whereby the machine learning model is trained based on a set of training examples and labels associated with the training examples. The weights may be stored on one or more computer-readable media, and are used by a system when applying the machine learning model to new data.

(67) The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the patent rights be limited not by this detailed description, but rather by any claims that issue on an application based hereon.

Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the patent rights, which is set forth in the following claims.

(68) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive “or” and not to an exclusive “or”. For example, a condition “A or B” is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present). Similarly, a condition “A, B, or C” is satisfied by any combination of A, B, and C having at least one element in the combination that is true (or present). As a not-limiting example, the condition “A, B, or C” is satisfied by A and B are true (or present) and C is false (or not present). Similarly, as another not-limiting example, the condition “A, B, or C” is satisfied by A is true (or present) and B and C are false (or not present).

# Claims

1. A method comprising: at a computer system, comprising at least one processor and memory: capturing an image by a camera coupled to a shopping cart, wherein the image depicts items on display in a region of a retailer location associated with a retailer; determining a location of the shopping cart within the retailer location when the image was captured by the shopping cart by capturing a location measurement using a location sensor coupled to the shopping cart; identifying, by a first process, an expected set of items for the image based on the captured image and the determined location, wherein the expected set of items is a set of items that are expected to be depicted in the image, and wherein identifying the expected set of items comprises: applying a first machine-learning model to the captured image to identify a set of barcodes depicted in the captured image, wherein the set of barcodes are coupled to a display for items depicted in the image; and identifying an item of a plurality of items associated with the retailer corresponding to each of the identified barcodes as an expected item for the expected set of items; identifying, by a second process separate from the first process, an actual set of items depicted in the image by applying a second machine-learning model to the image, wherein the second machine-learning model is computer-vision model that is trained to identify items depicted in an image, wherein the actual set of items is a set of items that are identified by the second machine-learning model as depicted in the image, wherein the expected set of items identified based on the identified barcodes comprises more items than the actual set of items depicted in the image; identifying a difference set of items, wherein the difference set of items comprises items that are in the expected set of items expected to be depicted in the image but are not in the actual set of items identified by the second machine-learning model as depicted in the image; updating an availability database for an online system to indicate that items in the difference set of items are unavailable; receiving an order from a client device associated with a user of the online system, wherein the order comprises an item from the difference set of items; determining that the item is unavailable by accessing the updated availability database; and transmitting, to the client device, an indication that the item is unavailable.
2. The method of claim 1, further comprising: receiving a search query request from a client device of a user, wherein the search query request comprises a search query; identifying a set of candidate items that are related to the search query from the search query request; and determining whether each candidate item in the set of candidate items is available by accessing the updated availability database.
3. The method of claim 2, further comprising: responsive to determining that a candidate item in the set of candidate items is unavailable, filtering the set of candidate items to remove the unavailable candidate item; and transmitting the filtered set of candidate items to the client device for presentation to the user.
4. The method of claim 3, further comprising: responsive to determining that a candidate item in the set of candidate items is unavailable, transmitting the set of candidate items to the client device for presentation to the user such that the unavailable candidate item is presented to the user with an indication that the unavailable candidate item is unavailable.
5. The method of claim 1, wherein determining the location of the shopping cart comprises: identifying a location marker in the image.
6. The method of claim 5, wherein the location marker is a QR code.
7. The method of claim 5, wherein the location marker is a barcode for an item in the expected set of items.
8. The method of claim 1, wherein identifying the expected set of items comprises: accessing an item map associated with the retailer location, wherein the item map indicates physical locations within the retailer location of items offered for collection within the retailer location.

9. The method of claim 1, wherein the location measurement of the shopping cart comprises an orientation of the shopping cart.

10. The method of claim 1, further comprising: transmitting a notification to a computing system operated by the retailer, wherein the notification indicates that the difference set of items are unavailable.

11. The method of claim 1, further comprising: training an item availability model based on the updated availability database.

12. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to: capture an image by a camera coupled to a shopping cart, wherein the image depicts items on display in a region of a retailer location associated with a retailer; determine a location of the shopping cart within the retailer location when the image was captured by the shopping cart by capturing a location measurement using a location sensor coupled to the shopping cart; identify, by a first process, an expected set of items for the image based on the captured image and the determined location, wherein the expected set of items is a set of items that are expected to be depicted in the image, and wherein identifying the expected set of items further comprises: applying a first machine-learning model to the captured image to identify a set of barcodes depicted in the captured image, wherein the set of barcodes are coupled to a display for items depicted in the image; and identifying an item of a plurality of items associated with the retailer corresponding to each of the identified barcodes as an expected item for the expected set of items; identify, by a second process separate from the first process, an actual set of items depicted in the image by applying a second machine-learning model to the image, wherein the second machine-learning model is a computer-vision model that is trained to identify items depicted in an image, wherein the actual set of items is a set of items that are identified by the second machine-learning model as depicted in the image, wherein the expected set of items identified based on the identified barcodes comprises more items than the actual set of items depicted in the image; identify a difference set of items, wherein the difference set of items comprises items that are in the expected set of items expected to be depicted in the image but are not in the actual set of items identified by the second machine-learning model as depicted in the image; update an availability database for an online system to indicate that items in the difference set of items are unavailable; receive an order from a client device associated with a user of the online system, wherein the order comprises an item from the difference set of items; determine that the item is unavailable by accessing the updated availability database; and transmit, to the client device, an indication that the item is unavailable.

13. The computer-readable medium of claim 12, further storing instructions that cause the processor to: receive a search query request from a client device of a user, wherein the search query request comprises a search query; identify a set of candidate items that are related to the search query from the search query request; and determine whether each candidate item in the set of candidate items is available by accessing the updated availability database.

14. The computer-readable medium of claim 13, further storing instructions that cause the processor to: responsive to determining that a candidate item in the set of candidate items is unavailable, filter the set of candidate items to remove the unavailable candidate item; and transmit the filtered set of candidate items to the client device for presentation to the user.

15. The computer-readable medium of claim 14, further storing instructions that cause the processor to: responsive to determining that a candidate item in the set of candidate items is unavailable, transmit the set of candidate items to the client device for presentation to the user such that the unavailable candidate item is presented to the user with an indication that the unavailable candidate item is unavailable.

16. The computer-readable medium of claim 12, wherein the instructions for determining the location of the shopping cart further cause the processor to: identify a location marker in the image.

17. The computer-readable medium of claim 12, wherein the instructions for identifying the



expected set of items further cause the processor to: access an item map associated with the retailer location, wherein the item map indicates physical locations within the retailer location of items offered for collection within the retailer location.

18. The computer-readable medium of claim 12, wherein the location measurement of the shopping cart comprises an orientation of the shopping cart.

19. A system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: capture an image by a camera coupled to a shopping cart wherein the image depicts items on display in a region of a retailer location associated with a retailer; determine a location of the shopping cart within the retailer location when the image was captured by the shopping cart by capturing a location measurement using a location sensor coupled to the shopping cart; identify, by a first process, an expected set of items for the image based on the captured image and the determined location, wherein the expected set of items is a set of items that are expected to be depicted in the image, and wherein identifying the expected set of items further comprises: applying a first machine-learning model to the captured image to identify a set of barcodes depicted in the captured image, wherein the set of barcodes are coupled to a display for items depicted in the image; and identifying an item of a plurality of items associated with the retailer corresponding to each of the identified barcodes as an expected item for the expected set of items; identify, by a second process separate from the first process, an actual set of items depicted in the image by applying a second machine-learning model to the image, wherein the second machine-learning model is a computer-vision model that is trained to identify items depicted in an image, wherein the actual set of items is a set of items that are identified by the second machine-learning model as depicted in the image, wherein the expected set of items identified based on the identified barcodes comprises more items than the actual set of items depicted in the image; identify a difference set of items, wherein the difference set of items comprises items that are in the expected set of items expected to be depicted in the image but are not in the actual set of items identified by the second machine-learning model as depicted in the image; update an availability database for an online system to indicate that items in the difference set of items are unavailable; receive an order from a client device associated with a user of the online system, wherein the order comprises an item from the difference set of items; determine that the item is unavailable by accessing the updated availability database; and transmit, to the client device, an indication that the item is unavailable.

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